CRICKET BALL TRACKING ANALYZER

Mini Project - II

Submitted in partial fulfilment of the academic requirement for the award of the degree of

BACHELOR OF ENGINEERING

In

ELECTRONICS AND COMMUNICATION ENGINEERING

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ABSTRACT

This project focuses on observing the tracking of the cricket ball. The goal of the project is to detect the cricket ball, capture its trajectory using various image processing techniques. By tracking the trajectory, we could extract various metrics such as speed, angle, and distance during the game. This helps to provide better insights into the bowler's performance.

The system takes the recorded videos and applies computer vision (OpenCV) techniques to detect the ball and track it. The system uses" ROBOFLOW MODEL" for ball detection and "YOLO V8 MODEL" for accurate ball tracking. After getting the trajectory, the system provides essential metrics like Ball swing intensity index and many other basic metrics during tracking. The results are displayed at the top left of the video for better understanding.

Overall, the project is useful for aspiring bowlers, analysts and many other cricket enthusiasts all over, who are keen towards the ball movements. It aims to reduce the time complexity by avoiding manual tracking. Ultimately, this project makes a small difference in the better understanding of the beautiful sport.

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CHAPTER – 1

INTRODUCTION

1.1 Introduction

In this modern world, sports are closely connected to technology for better analysis, performance and decision making. Cricket Ball Tracking Analyser is a computer-based (Open-CV) project built to detect the ball accurately without any noise and to track it during the game. Using Kaggle Datasets, the Roboflow model has been trained for accurate detection of the ball in different conditions. The system uses a Convolutional Neural Network to analyse the video frames, uses various filters for perfect ball tracking and provides different metrics like speed, distance, angle, and BSI. This helps the cricket coaches, bowlers, and analysts to shape themselves perfectly and make smarter decisions according to different conditions.

1.2 Aim

To analyse how the cricket ball moves after it is bowled, to know about its speed, swing, and angle using "YOLO V8 MODEL" for ball tracking, in order to give accuracy based on different lighting, pitch conditions.

1.3 Objectives

- To automatically detect the ball in every frame
- To track the ball's movement using coordinates
- To plot the trajectory and to visualise the path of the ball
- To extract basic metrics like angle, average speed, distance and advanced metrics like BSI (Ball Swing Intensity Index)

1.4 Methodology

1. Collect various sets of ball images and train them.
2. Input video of cricket bowling
3. YOLO V8 detects the ball in each frame
4. The tracking algorithm stores the ball in x-y co-ordinates
5. Calculate trajectory, extract metrics
6. Visualise results on graphs of the video output.

CHAPTER -2

LITERATURE REVIEW

Various researchers across the world have explored the usage of computer vision to analyse the movement of sports, especially in cricket. During the early times, they focused mainly on manual ball tracking, which is sometimes inaccurate and is highly time-consuming. With the advancement in technology, automated tracking systems gained attention and popularity. These systems helped to reduce human errors and gave accurate insights into the game.

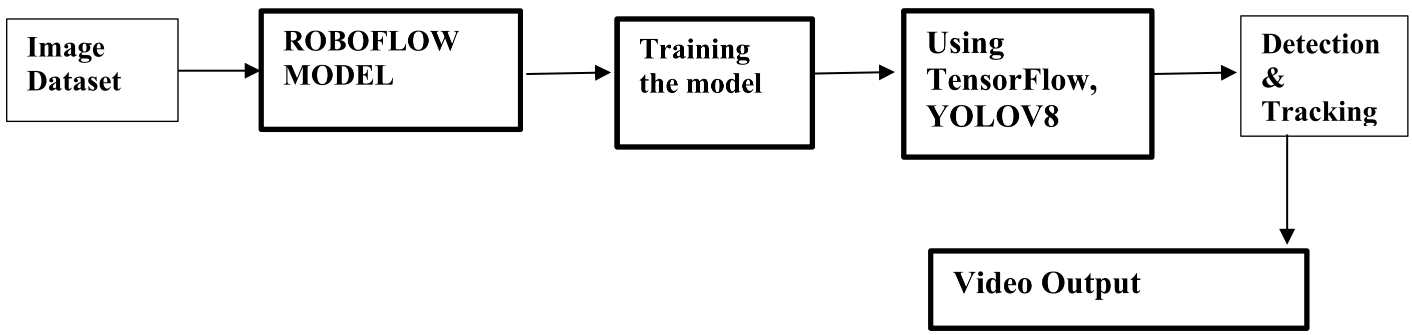
The technology has introduced Machine Learning techniques for the detection of the ball and motion tracking. Convolutional Neural Networks (CNNs) and various object detection models like YOLO have proved in identifying and detecting objects such as cricket balls, animal faces, and many more. Various types of datasets are available to train these models in different background conditions and pitch conditions.

Overall, developing an automated tracking system has become achievable and accurate for the analysts, researchers and aspiring bowlers who are about to bring a change in the wonderful sport. These advancements in technology reduced human errors, avoided manual tracking and improved decision–making smartly.

CHAPTER 3

Block Diagram/Flow Diagram

3.1 Block Diagram



3.2 Explanation

- A set of data images was collected from various sources.
- The collected images are sent to the **ROBOFLOW** model for bounding boxes.
- The annotated dataset is used for training the required model.
- The **YOLOv8** model is used to detect the ball in any type of input conditions.
- The final output is when the ball gets detected perfectly, and the trajectory will be displayed.

CHAPTER 4

Project Implementation

4.1 System Modules

- **Dataset Preparation Module –** collects images of cricket balls
- **ROBOFLOW Training Module** – to annotate the ball and provide bounding boxes
- **YOLO V8 Model** – Object detection model
- **Tracking Module –** for continuous tracking of the ball.
- **Output Module –** output video is displayed with metrics

```
import cv2
import numpy as np
from ultralytics import YOLO
import matplotlib.pyplot as plt
from collections import deque
import math
```

```
from roboflow import Roboflow
from ultralytics import YOLO
import multiprocessing
```

4.1 Module Imports

4.2 Core Algorithm

A. Object Detection

YOLOv8 processes each frame in real time, enabling the system to detect multiple objects with high accuracy and low delay.

B. Tracking Module

It continuously follows the ball movement using a deque to store the previous ball positions.

C. Kalman Filter

It is a mathematical algorithm used to predict the future position of the ball based on its present position

D. Ball Swing Intensity Index

A metric used to measure how a cricket ball deviates sideways.

4.3 Cricket Ball Detection using YOLOv8

```
def main():
    rf = Roboflow(api_key="BtNwCe4uvuGwoWjvQnZS")
    project = rf.workspace().project("cricket_ball_detection-3ukl0")
    dataset = project.version(1).download("yolov8")
    model = YOLO("yolov8n.pt")
    dataset_path = dataset.location

    model.train(
        data=f"{dataset_path}/data.yaml",
        epochs=100,
        imgsz=640,
    )

    results = model.predict(source=r"C:\Users\vivek\Downloads\ms-dhoni-practice-1559564708.jpg",
    print(results)
```

4.4 Kalman Filter

```python
def setup_kalman_filter(self):
    """Initialize Kalman filter for smooth trajectory prediction"""
    kalman = cv2.KalmanFilter(4, 2)

    # Transition matrix (x, y, vx, vy)
    kalman.transitionMatrix = np.array([
        [1, 0, 1, 0],
        [0, 1, 0, 1],
        [0, 0, 1, 0],
        [0, 0, 0, 1]
    ], np.float32)

    # Measurement matrix
    kalman.measurementMatrix = np.array([
        [1, 0, 0, 0],
        [0, 1, 0, 0]
    ], np.float32)

    # Process noise covariance
    kalman.processNoiseCov = np.eye(4, dtype=np.float32) * 0.03

    # Measurement noise covariance
    kalman.measurementNoiseCov = np.eye(2, dtype=np.float32) * 0.1

    return kalman
```

4.5 Calculating Ball Swing Intensity Index

```python
def calculate_bsi(self, positions):
    """
    Calculate Ball Swing Intensity (BSI).We fit a straight line to the early segment (first 20% or at least 3 points),then compute
    if len(positions) < 3:
        return 0.0

    pts = np.array(positions, dtype=float)
    x = pts[:, 0]
    y = pts[:, 1]

    # choose reference segment: first 20% of points (at least 3)
    n_ref = max(3, len(pts) // 5)
    x_ref = x[:n_ref]
    y_ref = y[:n_ref]

    # Fit line y = m*x + c (least squares). If vertical-ish, fit x = ay + b instead.
    # Use exception handling for degenerate cases.
    try:
        m, c = np.polyfit(x_ref, y_ref, 1)  # slope m and intercept c
        # perpendicular distance formula for line y = m x + c
        denom = math.sqrt(m*m + 1.0)
        dists = np.abs(m * x - y + c) / denom
    except Exception:
        # fallback: if polyfit fails (near-vertical), fit x = a*y + b
        a, b = np.polyfit(y_ref, x_ref, 1)
        denom = math.sqrt(a*a + 1.0)
        dists = np.abs(a * y - x + b) / denom

    bsi_value = float(np.mean(dists))
    return round(bsi_value, 3)
```

4.6 Drawing Trajectory

```python
def draw_trajectory(self, frame):
    """Draw ball trajectory on the frame"""
    if len(self.trajectory) < 2:
        return frame

    # Draw trajectory path with gradient color
    for i in range(1, len(self.trajectory)):
        intensity = int(255 * (i / max(1, len(self.trajectory)-1)))
        color = (0, intensity, 255 - intensity)  # Blue to Red gradient
        thickness = max(2, int(3 * (i / max(1, len(self.trajectory)-1))))
        cv2.line(frame, self.trajectory[i-1], self.trajectory[i], color, thickness)

    # Draw current position
    if self.trajectory:
        current_pos = self.trajectory[-1]
        cv2.circle(frame, current_pos, 8, (0, 255, 0), -1)  # Green dot
        cv2.circle(frame, current_pos, 8, (255, 255, 255), 2)  # White border

    return frame
```

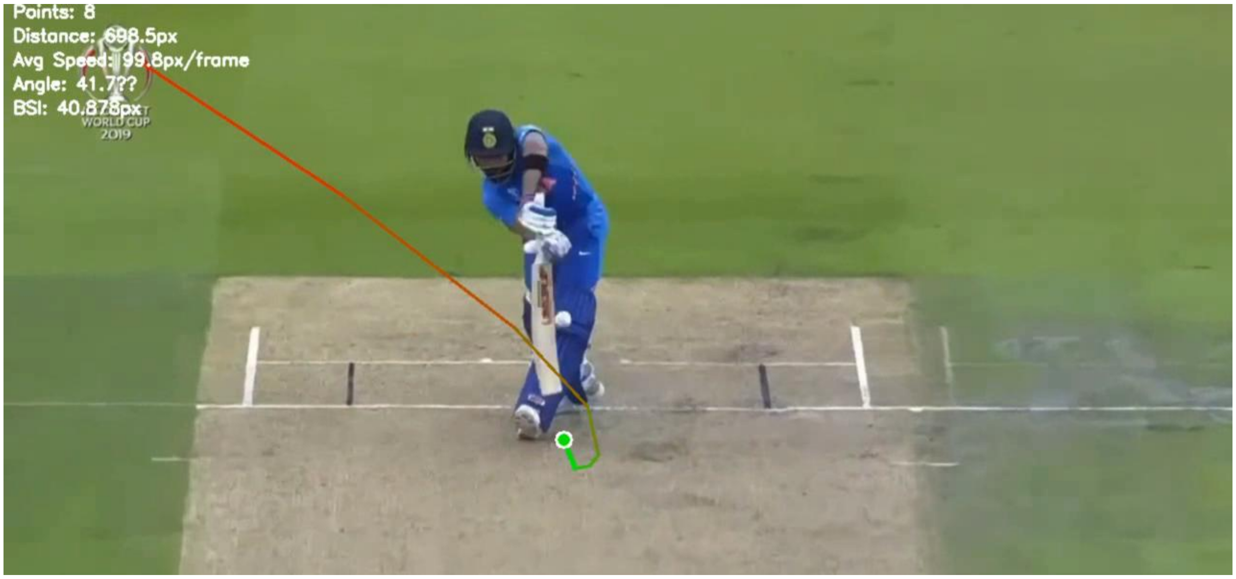
4.7 Displaying the metrics

```python
def draw_detection_info(self, frame, detection, metrics):
    """Draw detection information and metrics on frame"""
    if detection:
        x1, y1, x2, y2 = detection['bbox']
        confidence = detection['confidence']
        # Draw bounding box
        cv2.rectangle(frame, (int(x1), int(y1)), (int(x2), int(y2)), (0, 255, 0), 2)
        # Draw confidence label
        label = f"Ball: {confidence:.2f}"
        cv2.putText(frame, label, (int(x1), int(y1) - 10),
                    cv2.FONT_HERSHEY_SIMPLEX, 0.6, (0, 255, 0), 2)

    # Draw trajectory metrics
    if metrics:
        y_offset = 30
        metrics_text = [
            f"Points: {metrics.get('total_points', 0)}",
            f"Distance: {metrics.get('total_distance', 0.0):.1f}px",
            f"Avg Speed: {metrics.get('average_speed', 0.0):.1f}px/frame",
            f"Angle: {metrics.get('direction_angle', 0.0):.1f}°",
            f"BSI: {metrics.get('bsi', 0.0):.3f}px"
        ]

        for text in metrics_text:
            cv2.putText(frame, text, (10, y_offset),
                        cv2.FONT_HERSHEY_SIMPLEX, 0.6, (255, 255, 255), 2)
            y_offset += 25
        return frame
```

4.8 Terminal Output

```
PS C:\Users\vivek> python -u "c:\Users\vivek\Documents\Cricket Ball Analyzer\Cricket-Ball-Trajectory-Prediction\trajectory.py"
Processing video: C:\Users\vivek\Documents\Cricket Ball Analyzer\Cricket-Ball-Trajectory-Prediction\videos\test3.mp4
Resolution: 1280x720, FPS: 30, Total Frames: 89
Processing completed! Output saved to: C:\Users\vivek\Documents\Cricket Ball Analyzer\Cricket-Ball-Trajectory-Prediction\output\output.mp4
Trajectory analysis plot saved to: C:\Users\vivek\Documents\Cricket Ball Analyzer\Cricket-Ball-Trajectory-Prediction\output\output.png

=== FINAL TRAJECTORY STATISTICS ===
Total trajectory points: 26
Total distance covered: 1274.69 pixels
Average speed: 50.99 pixels/frame
Direction angle: 18.5°
BSI: 68.172 px (average perpendicular deviation)
Start point: (146, 79)
End point: (936, 344)
```

4.9 Output Snippets



4.10 Software Dependencies

The following are the dependencies for this project:

- **Python:** Used as the base programming Language
- **OpenCV:** Video Capture and Preprocessing
- **YOLOv8n:** Real-time object detection

CHAPTER 4

Results

5.1 Functional Testing (Software-Level Verification)

The cricket ball tracking analyser has successfully and accurately detected the cricket ball image regardless of any spherical image given. The dataset was trained using an annotated set in the Roboflow model, which can receive and recognise a cricket ball with accuracy and precision under any lighting conditions. We were able to draw a bounding box in each and every frame possible, smoothly track the ball using the Kalman Filter and X-Y coordinates, even when the ball speed is high, or the ball is motion-blurred. We were able to generate a trajectory plot and display various metrics like swing, speed, distance, angle and advanced metrics like Ball Swing Intensity Index. Finally, the output graphs have been plotted according to the given input video.

Conclusion:

The cricket ball tracking has been demonstrated using the Roboflow and YOLOv8 model, which has the capability to switch to real-time analysis in further development. The system was able to accurately track and locate the cricket ball from the initial frame to the last frame of the video. The inclusion of the Kalman Filter smoothes the trajectory and is able to track even if the ball is out of sight; it is able to predict the next position based on the present position. This is a small foundation for local cricket tournaments, mainly used for coaches, aspiring bowlers and cricket enthusiasts globally.

Future Scope:

1. We can switch to real-time tracking.
2. We can include various advanced metrics like bounce prediction and player action recognition.
3. It could be utilised for the 3-D trajectory to increase the precision.
4. We can make a mobile application suitable for coaches and players during practice sessions.

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